

The background of the slide is a dark blue field filled with a complex network of glowing blue circles and lines, resembling a molecular structure or a network diagram. The circles vary in size and brightness, with some appearing as bright, out-of-focus light sources. The lines connect these circles, creating a web-like pattern that suggests chemical bonds or data connections.

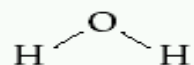
Lec(6):Valence Shell Electron- Pairs Repulsion Theory "VSEPR"

Molecular Geometry and Chemical Bonding Theories

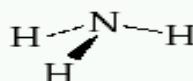
Valence Shell Electron-Pairs Repulsion Theory "VSEPR"

- It is a method for predicting the shape of a molecule from the knowledge of the groups of electrons around a central atom.
- VSEPR is revolves around the principle that electrons repel each other.
- One can predict the shape of a molecule by finding a pattern where electron pairs are as far from each other as possible.

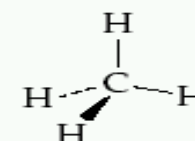
- The shape of a molecule has an important bearing on its reactivity and behavior.
- The shape of a molecule depends on a number of factors. These include:
 1. Atoms forming the bonds
 2. Bond distance



Water, H_2O



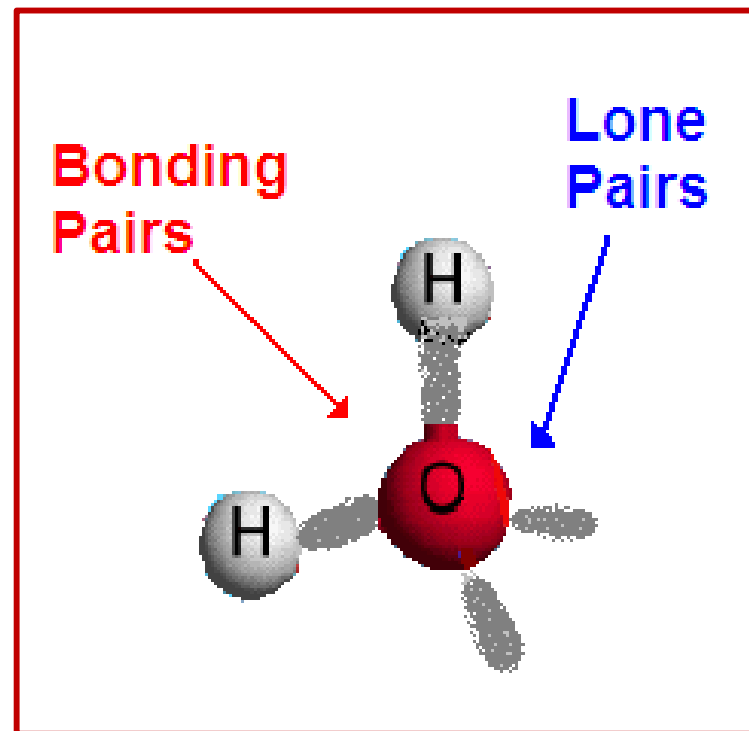
Ammonia, NH_3



Methane, CH_4

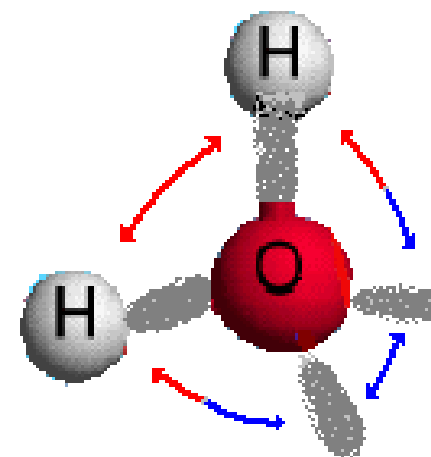
Bonding Electrons and Lone Pairs

- In a molecule some of the valence electrons are shared between atoms to form covalent bonds. These are called **bonding electrons**.
- Other valence electrons may not be shared with other atoms. These are called **non-bonding electrons** or they are often referred to as **lone pairs**.



VSEPR

- In all covalent molecules, electrons will tend to stay as far away from each other as possible
- The shape of a molecule therefore depends on:
 1. the number of regions of electron density it has on its central atom,
 2. whether these are bonding or non-bonding electrons.



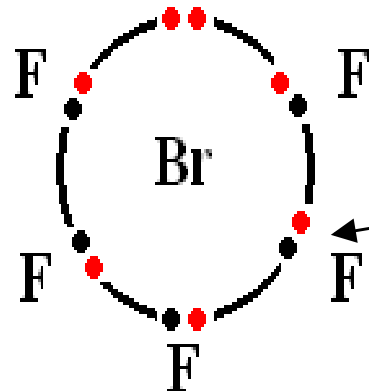
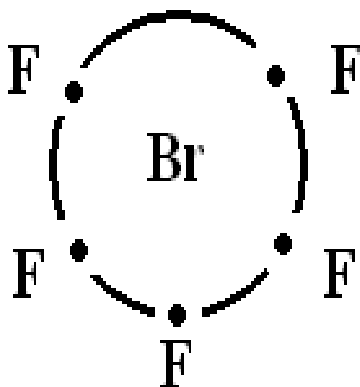
Both bonding and non-bonding electron pairs repel

Applying the VSEPR theory

In order to use VSEPR to predict molecular structure:

- 1) Draw up Lewis dot diagram for the molecule or ion. The first atom (e.g. Br in BrF_5) is always the central atom. Place the other atoms around the central atom.

place 5 F
atoms around
central Br



red = 7 valence
electrons for Br

If these are single bonds, contribute one electron per attached atom. Then add the valence electrons for the central atom = 7 for Br.

2) Work out number of electron domains = valence electron pairs ('n') plus attached atoms on central atom. For BrF_5 $n = 6$.

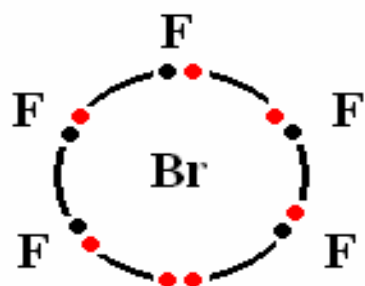
3) Relate n to the type of structure predicted for that value of n. $n = 6 = \text{octahedral}$.

4) Place lone pairs in expected positions, maximizing separation of lone pairs. For BrF_5 , there is one lone pair, so



The structure of BrF_5 from VSEPR:

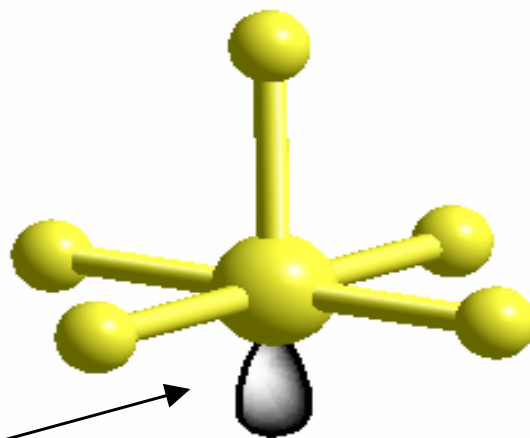
Lewis dot diagram



lone pair

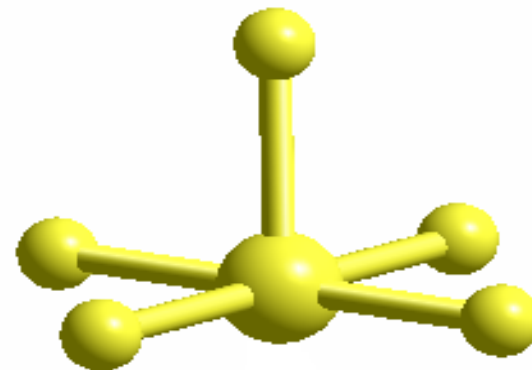
$n = 6$ from five attached atoms plus one electron pair

parent structure



$n = 6$, parent structure = octahedral, but one site occupied by a lone pair

molecular structure = square pyramidal



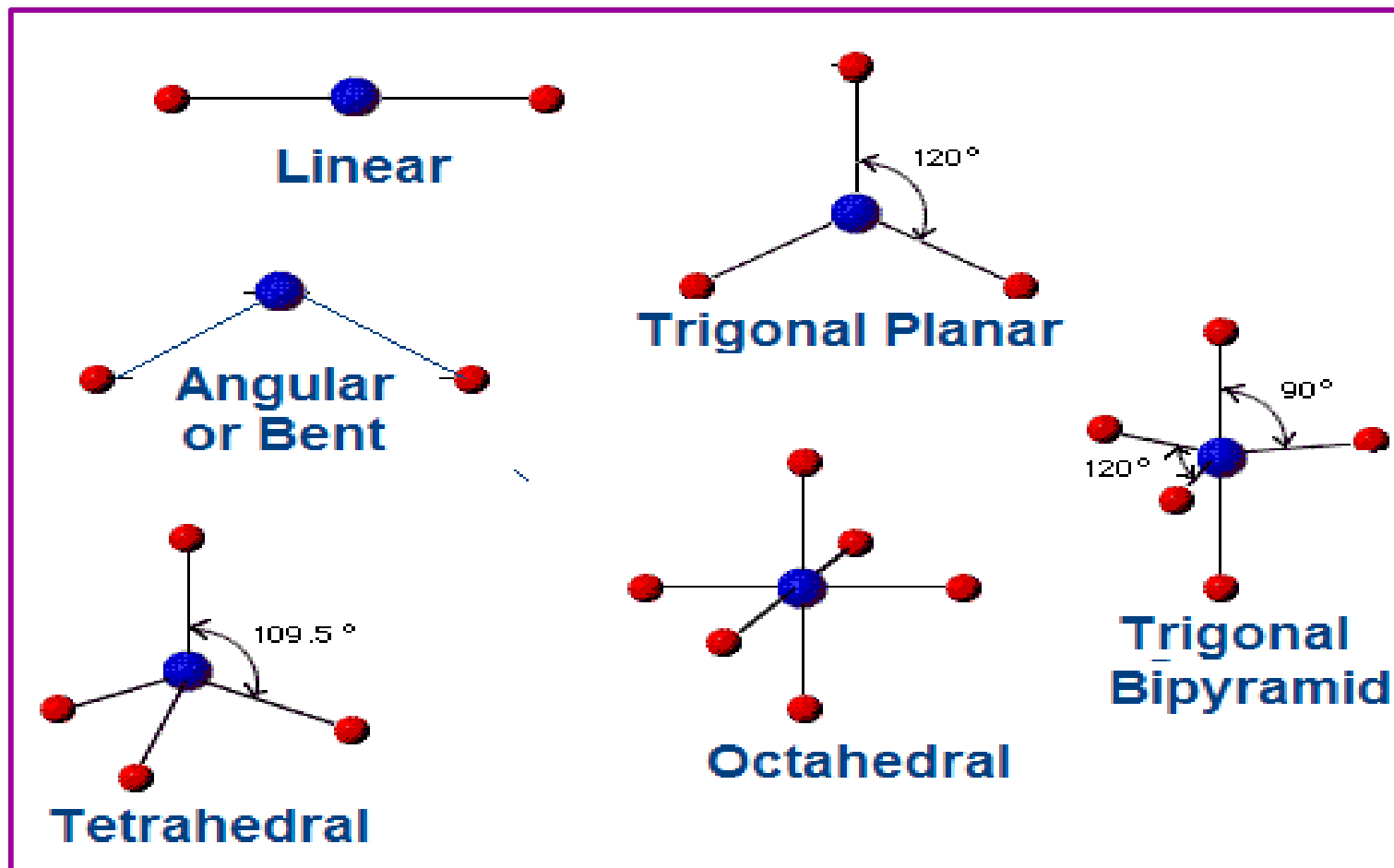
molecular or final structure – disregard the lone pair

VSEPR: Predicting the shape

- Once the dot structure has been established, the shape of the molecule will follow one of basic shapes depending on:
 1. The number of regions of electron density around the central atom.
 2. The number of regions of electron density that are occupied by bonding electrons.
- The number of regions of electron density around the central atom determines the **electron skeleton**
- The number of regions of electron density that are occupied by bonding electrons and hence other atoms determines the **actual shape**.

Basic Molecular Shapes

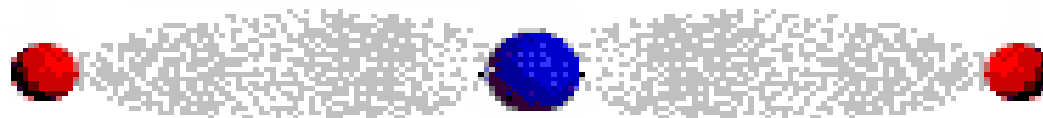
The most common shapes of molecules are shown below:



Linear Molecules

Linear molecules have only two regions of electron density.

Linear



There are 2 regions of electron density around the central atom. Both are occupied by other atoms

Examples



Angular or Bent

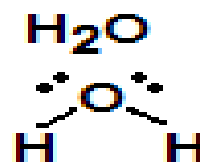
Angular or bent molecules have at least 3 regions of electron density, but only two are occupied.

Angular or Bent

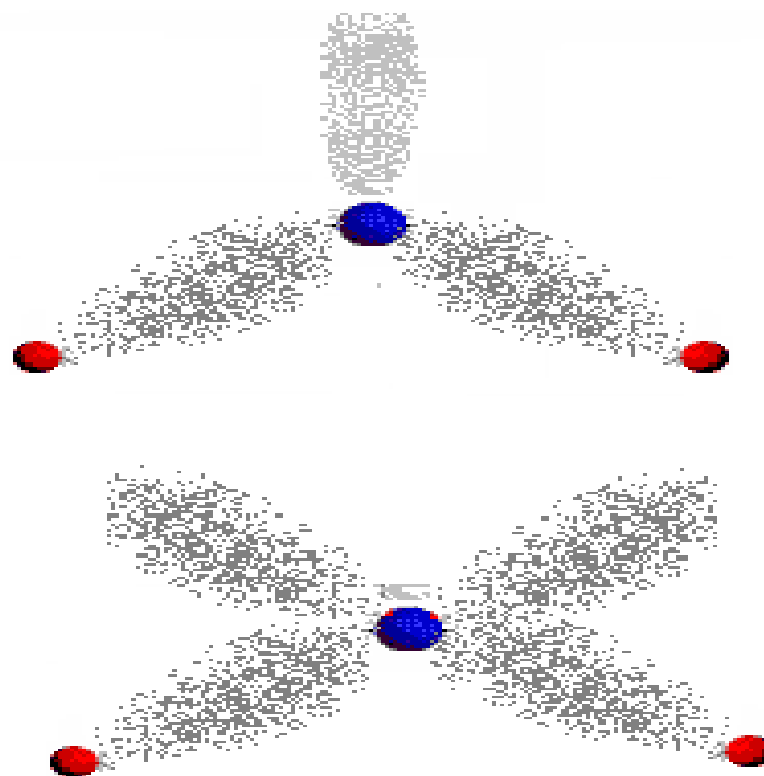
Example



1 lone pair



2 lone pairs



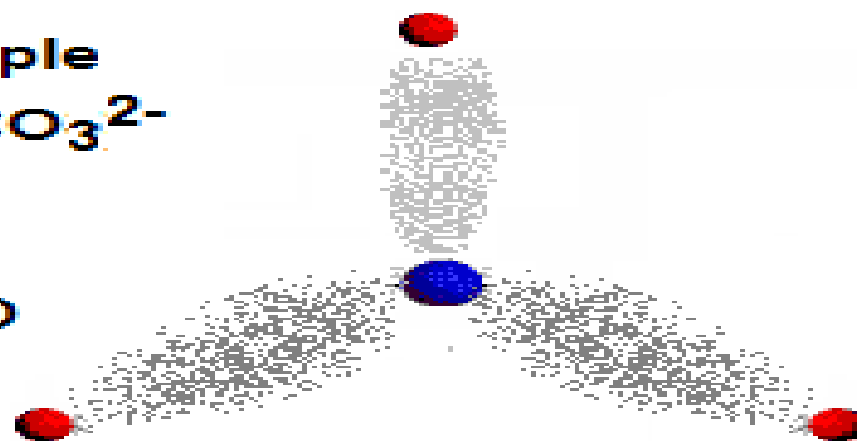
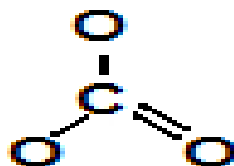
Angular or bent molecules have 3 or more regions of electron density but only two regions are bonded to other atoms

Triangular Planar

Triangular planar molecules have three regions of electron density. All are occupied by other atoms.

Triangular Plane

Example



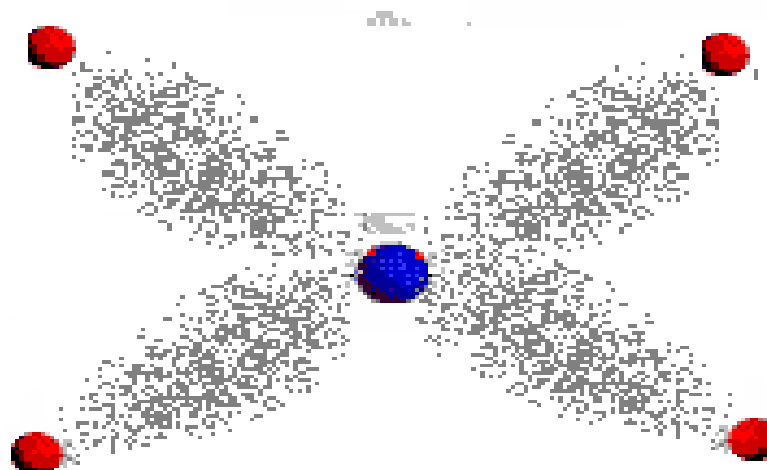
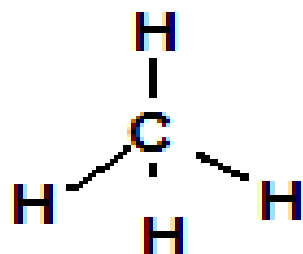
Triangular planar molecule have three regions of electron density. All are occupied by other atoms. The resulting shape is a triangular plane.

Tetrahedron

Tetrahedral molecules have four regions of electron density. All are occupied by other atoms:

Tetrahedron

Example



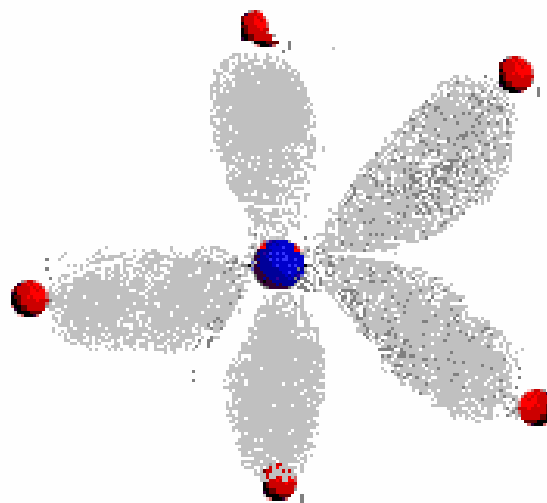
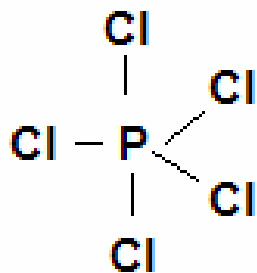
A tetrahedral molecule has four regions of electron density. All are occupied by other atoms. The result is a three dimensional figure with bond angles of about 109°

Triagonal Bipyramid

A few molecules have **expanded valence shells** around the central atom. Hence there are **five pairs of valence electrons**. The structure of such molecules with five pairs around one is called trigonal bipyramid.

Trigonal Bipyramid

Example
 PCl_5



In a trigonal bipyramid the central atom is surrounded by 5 pairs of bonded electrons. The valence shell around the central atom is expanded to ten electrons instead of the usual eight.

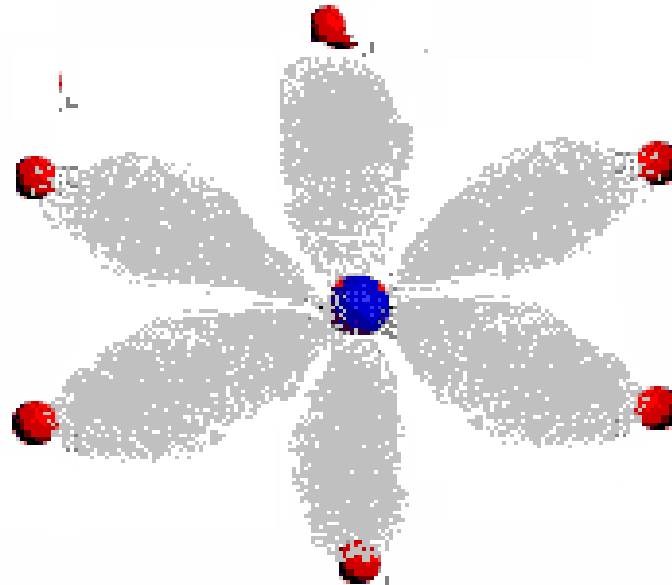
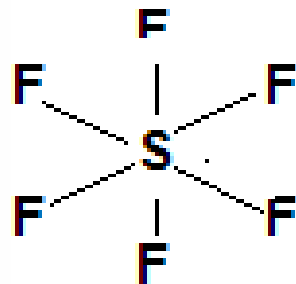
Octahedron

A molecules have valence shells around the central atom that are expanded to as many as **six pairs or twelve electrons**. These shapes are known as **octahedrons**.

Octahedron

Example

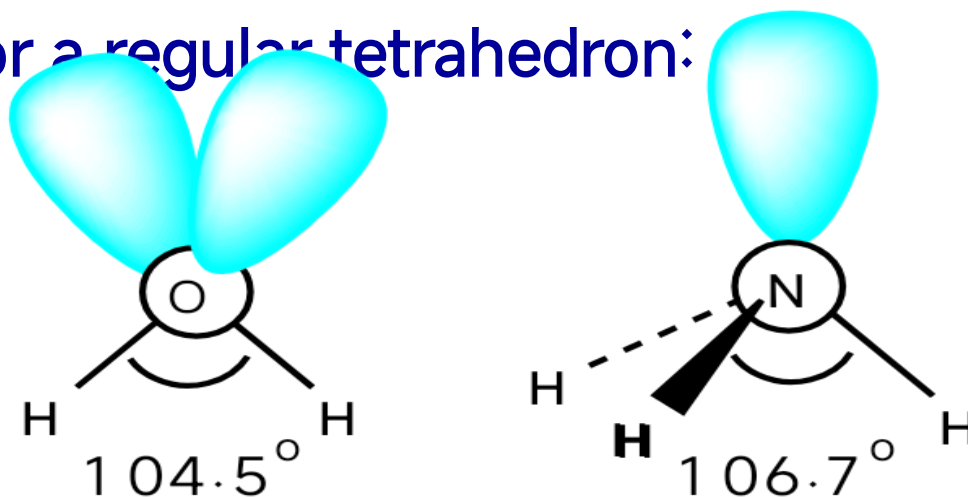
SF_6



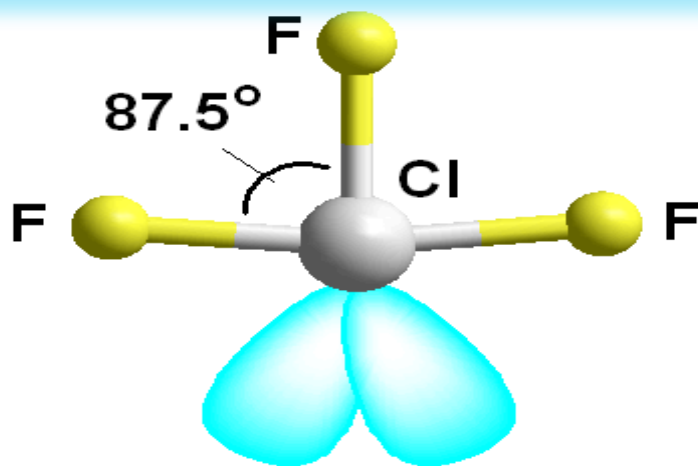
An octahedral molecule has **six regions of electron density** around the central atom. The valence shell is expanded to **12 electrons** instead of the usual eight

The effect of lone pairs on bond angles

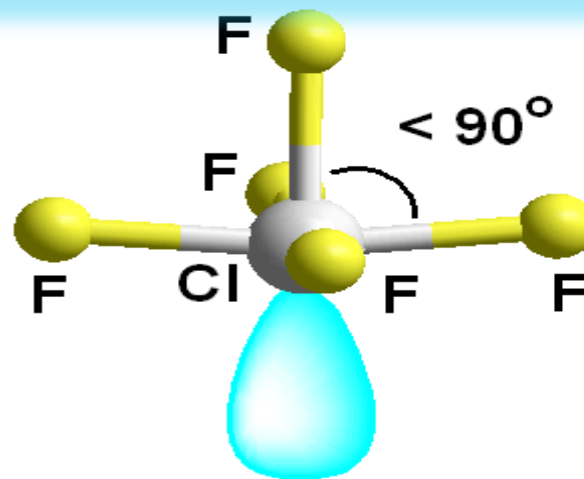
In VSEPR the lone pairs appear to occupy more space than electron pairs in bonds, with the result that bond angles are compressed away from the lone pairs. For example, in structures derived from tetrahedral parent geometry, such as water or ammonia, the H-O-H and H-N-H angles are compressed to be less than the 109.5° expected for a regular tetrahedron:



Effects of lone pairs on bond angles in ClF_3 and ClF_5



chlorine trifluoride



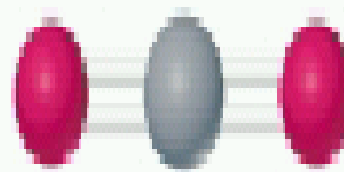
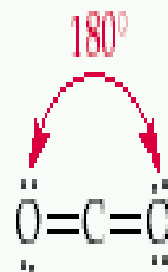
chlorine pentafluoride

Lone pairs tend to compress the other bond angles in a molecule, because the repulsion increases in the

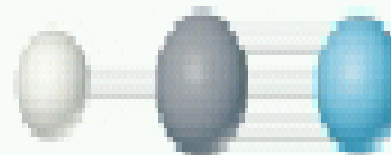
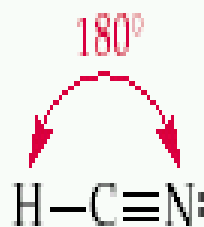
order; Two Lone Pairs > Lone Pair and Bond Pair > Two Bond Pairs

Question: predict the shape of CO_2 , HCN , CH_4 , NH_3 , SO_2 , PCl_5 , SF_6 and H_2O by using VSEPR theory.

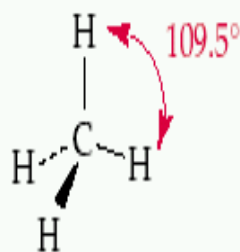
A CO_2 molecule
is linear, with a bond
angle of 180° .



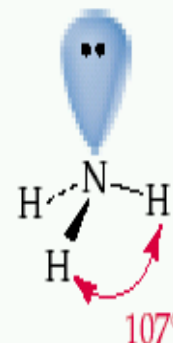
An HCN molecule
is linear, with a bond
angle of 180° .



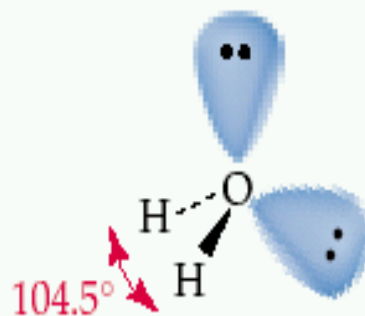
A methane molecule is tetrahedral, with bond angles of 109.5° .

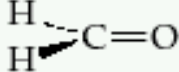
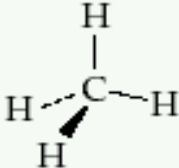
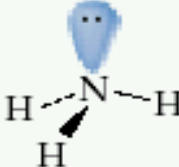
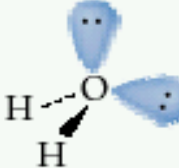


An ammonia molecule is trigonal pyramidal, with bond angles of 107° .

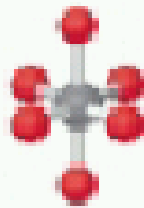
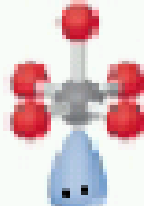


A water molecule is bent, with a bond angle of 104.5° .



Number of Bonds	Number of Lone Pairs	Number of Charge Clouds	Molecular Geometry	Example
2	0	2	 Linear	$\text{O}=\text{C}=\text{O}$
3 2	0	3	 Trigonal planar	
	1		 Bent	
4 3 2	0	4	 Tetrahedral	
	1		 Trigonal pyramidal	
	2		 Bent	

Number of Bonds	Number of Lone Pairs	Number of Charge Clouds	Molecular Geometry	Example
5	0	5	Trigonal bipyramidal	
4	1		Seesaw	
3	2		T-shaped	
2	3		Linear	

Number of Bonds	Number of Lone Pairs	Number of Charge Clouds	Molecular Geometry	Example
6	0	6	Octahedral	
5	1		Square pyramidal	
4	2		Square planar	